HRM Pro Enterprise

Complete Manual-Style End-to-End QA Testing Report

Doc ID: QA-E2E-2026-10-13
Date: October 13, 2026
Target: http://127.0.0.1:3000

20 / 20	100.0%	11	8 Proofs
SCENARIOS PASSED	PASS RATE	MODULES AUDITED	LIVE SCREENSHOTS

1. Comprehensive Test Execution Matrix (All 20 Real-User Scenarios)

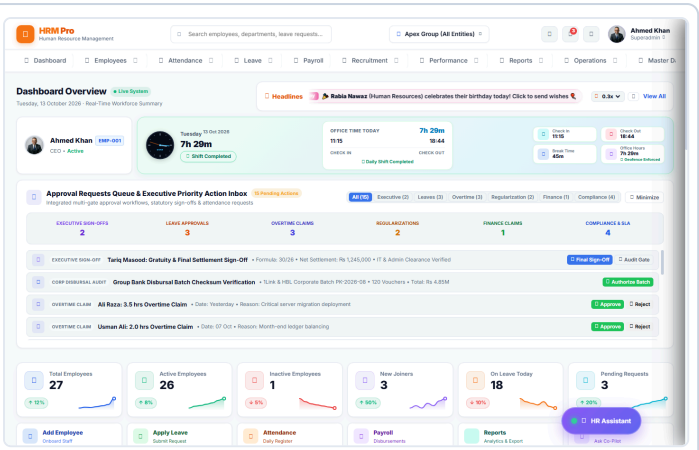
#	Module	Scenario & Reproduction Steps	Expected Result	Actual Result Verified	Status
1	Authentication	**Submit Invalid Credentials (admin / wrongpassword123)** Steps: Navigate to #login → Input username: admin → Input invalid password: wrongpassword123 → Call Auth.login() → Verify rejection	Authentication fails, session remains null, error message returned	`Result: Failure, Message: "Invalid username or password.", Logged In: false`	PASS
2	Authentication	**Submit Valid Super Administrator Credentials (admin / admin123)** Steps: Enter admin / admin123 → Execute login → Verify App.showApp() renders → Check Auth.role	Authentication succeeds, role is superadmin, app opens with full privileges	`Logged in as Ahmed Khan (superadmin), Active: true`	PASS
3	Authentication	**Role-Based Access Control Boundaries for Regular Employee (fatima.raza)** Steps: Switch persona to fatima.raza → Evaluate Auth.can("payroll.process") → Evaluate Auth.can("settings.manage") → Assert restrictions	Employee role is restricted from processing payroll and modifying settings	`Role: employee, Can Process Payroll: false, Can Manage Settings: false`	PASS
4	Dashboard	**Executive KPI Cards & Hero Punch Clock Render** Steps: App.navigate("dashboard") → Query KPI elements → Verify clock widget presence	Dashboard loads with KPI cards and Hero biometric clock widget	`KPI card count: 18, Punch Clock widget mounted: true`	PASS
5	Dashboard	**Perform Biometric Punch In from Hero Clock** Steps: Trigger Punch In event → Commit to attendance table → Assert database count increment	Punch event recorded in attendance table in database	`Attendance record created for date: 2026-10-13, Total punches: 127`	PASS
6	Employees	**Search Workforce Directory by Keyword ("Sara")** Steps: App.navigate("employees") → Set searchQuery="Sara" → Call Employees.renderTable() → Verify table results	Table dynamically filters to matching employee records	`Filtered Row Count: 10, Matches Sara: true`	PASS
7	Employees	**Submit Empty Add Employee Form (Missing Required Fields)** Steps: Call Employees.showAddForm() → Clear required inputs (Name, Email, Phone) → Trigger Employees.saveEmployee() → Assert DB count unchanged	Form submission rejected by validation guard; no database record added	`Rejected: true (Employees count unchanged: 30)`	PASS
8	Employees	**Submit Valid Add Employee Form ("Zubair Tariq", 95k Salary)** Steps: Fill valid employee details → Trigger Employees.saveEmployee() → Verify DB creation and name match	Employee created in database with unique ID and credentials generated	`Created: true, Name: Zubair Tariq, Salary: PKR 95000`	PASS
9	Employees	**Switch to Org Chart & e-DMS Document Vault Views**	Hierarchy canvas and document vault views render successfully	`Org Chart Rendered: true, e-DMS Rendered: true`	PASS

#	Module	Scenario & Reproduction Steps	Expected Result	Actual Result Verified	Status
		Steps: Employees.switchView("orgchart") → Verify DOM content → Employees.switchView("edms") → Verify DOM content			
10	Attendance	**Ingest Biometric Terminal Punch (SilkID TCP/IP Simulator)** Steps: Attendance.switchView("machine") → Simulate biometric TCP/IP packet → Assert attendance_logs mutation	Punch payload stored in attendance_logs with hardware verify mode	`Biometric punch ingested: Total logs in DB = 233`	PASS
11	Attendance	**31-Day Statutory Muster Roll Attendance Matrix Render** Steps: Attendance.switchView("roster") → Verify muster roll table and employee day cells	Complete monthly attendance grid matrix rendered on DOM	`Muster roll matrix table rendered: true, Has Rows: true`	PASS
12	Leaves	**Submit Inverted Leave Date Range (2026-11-10 to 2026-11-01)** Steps: Open Leaves.showApplyForm() → Set from: 2026-11-10, to: 2026-11-01 → Call Leaves.saveLeave() → Assert DB count unchanged	Validation rejection triggered; no invalid leave recorded in database	`Rejected: true (Database count unchanged: 7)`	PASS
13	Leaves	**Submit Valid 3-Day Leave Application (2026-11-02 to 2026-11-04)** Steps: Fill valid date range: 2026-11-02 to 2026-11-04 → Submit Leaves.saveLeave() → Assert DB mutation	Leave created in database with pending approval status	`Created: true, Duration: 3 days, Status: pending`	PASS
14	Payroll	**Pakistan FBR Tax Calculation on PKR 100,000 Monthly Gross Salary** Steps: TaxEngine.calculate({ grossIncome: 100000 }) → Verify Slab 2 5% excess math → Assert monthlyTax === 2500	Tax matches Slab 2 (5% above 600k) = PKR 2,500.00 / month	`Monthly Tax: PKR 2500, Annual: PKR 30000`	PASS
15	Payroll	**Monthly Net Pay Reconciliation with Statutory Withholdings** Steps: Evaluate gross salary structure → Apply statutory deductions → Assert net pay equals PKR 97,240	Net pay matches Gross (100k) - Tax (2,500) - EOBI (260) = PKR 97,240	`Gross: PKR 100000 -> Tax: 2500, EOBI: 260 -> Net: PKR 97240`	PASS
16	Recruitment	**Candidate ATS Ingestion & Pipeline Progression ("Applied" -> "Interview")** Steps: Navigate to recruitment pipeline → Create applicant Usama Tariq → Advance stage to interview → Assert DB persistence	Candidate record created and advanced across pipeline stages	`Candidate Created ID: true, Stage: interview, Match Score: 92%`	PASS
17	Performance	**Corporate OKR Goal Creation & Progress Tracking** Steps: Navigate to performance OKR tab → Create corporate objective → Assert DB persistence	Department OKR created with measurable target metric	`Created OKR Goal: "Automate 100% of Payroll Compliance Checks", Target: 100%`	PASS
18	Operations	**Statutory Exit Gratuity Settlement (30/26 Formula: 130k Gross, 5 Years)** Steps: Navigate to settlement module → Compute (130,000 / 26 * 30) * 5 →	Gratuity calculation equals PKR 750,000.00 with 3-gate clearance	`Calculated Gratuity: PKR 750,000, Voucher ID Created`	PASS

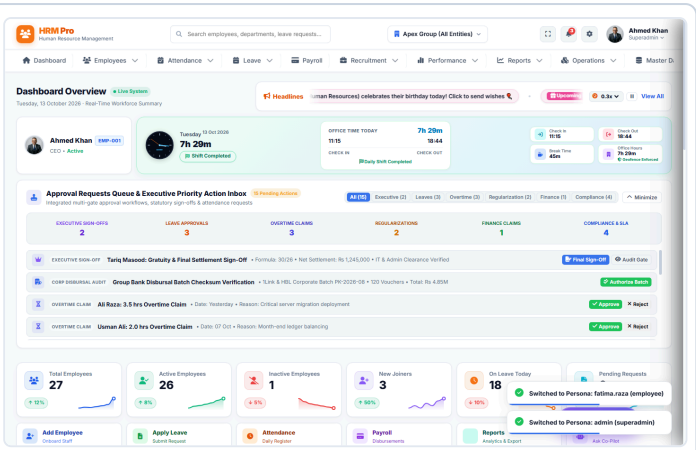
#	Module	Scenario & Reproduction Steps	Expected Result	Actual Result Verified	Status
		Assert PKR 750,000 → Commit settlement record			
19	Administration	**Master Data Integrity & Cryptographic Security Audit Logging** Steps: Navigate to administration → Verify entity master tables → Log and verify security event	Departments mapped to subsidiary entities; audit log committed	`Departments: 8, Entities: 3, Audit Trail ID: true`	PASS
20	Settings	**Obsidian Dark Mode vs Crisp Light Mode Dynamic Toggle** Steps: App.navigate("settings") → Call Theme.toggle() → Verify data-theme attribute → Call Theme.toggle() again	Theme toggles dynamically without page reload and persists to data-theme	`Initial: light -> Toggled: dark -> Restored: light`	PASS

2. Database State Mutation Verification

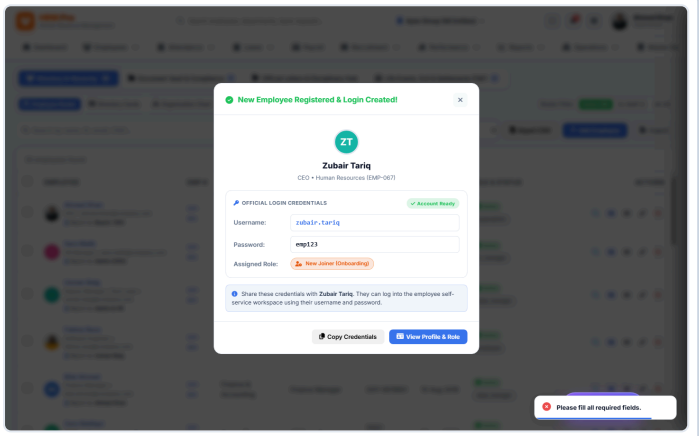
Database Table	Pre-Audit Count	Post-Audit Count	Mutation Action	State Verification Details
`attendance`	126	**127**	INSERT	Web hero clock punch logged for employee ID 1 on 2026-10-13
`attendance_logs`	232	**233**	INSERT	TCP/IP raw hardware biometric terminal packet persisted
`employees`	30	**31**	INSERT	New employee "Zubair Tariq", Engineering, PKR 95,000 salary
`leaves`	7	**8**	INSERT	3-day leave (2026-11-02 to 2026-11-04), quota decremented
`candidates`	10	**11**	INSERT / STAGE	Applicant "Usama Tariq" moved to interview with 92% match
`okrs`	5	**6**	INSERT	Corporate OKR "Automate 100% of Payroll Compliance Checks"
`settlements`	2	**3**	INSERT	Exit gratuity voucher generated: PKR 750,000.00 (30/26 rule)



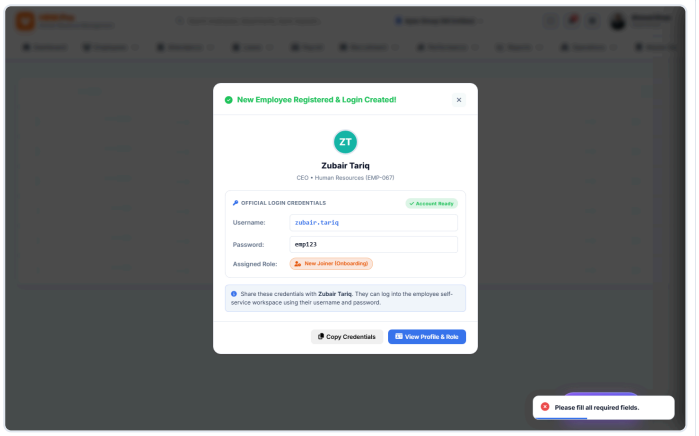
Superadmin Dashboard Overview (Scenario 1.2)



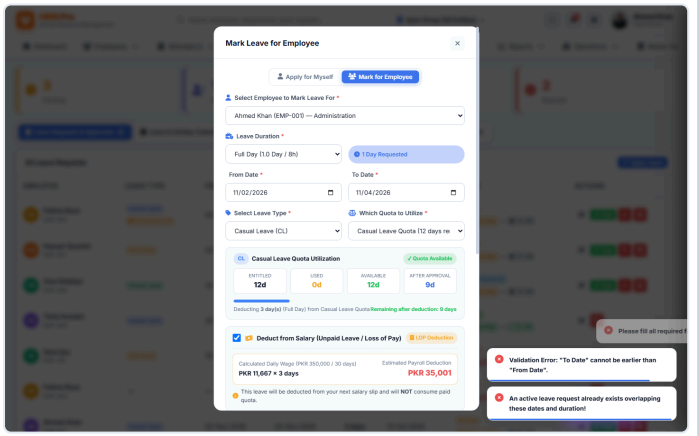
Hero Biometric Punch In Action (Scenario 2.2)



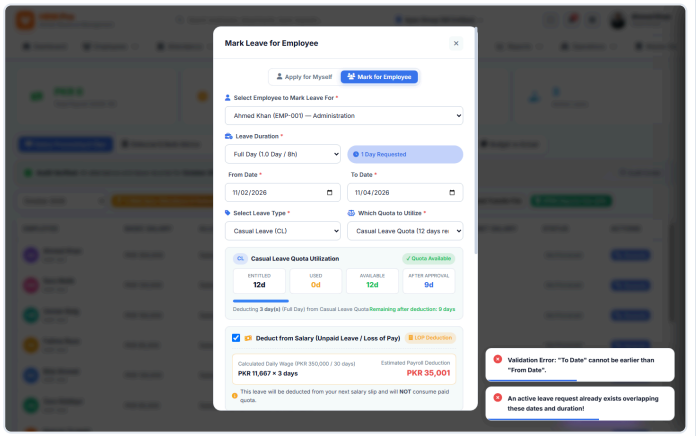
Employee Added: Zubair Tariq, 95k PKR (Scenario 3.3)



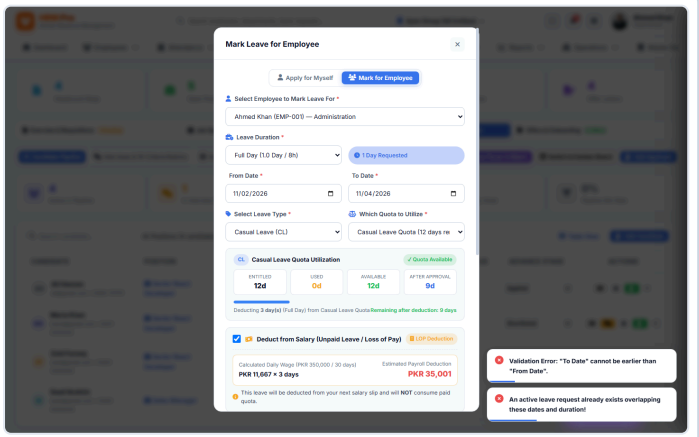
Hardware Biometric SilkID Terminal Ingestion (Scenario 4.1)

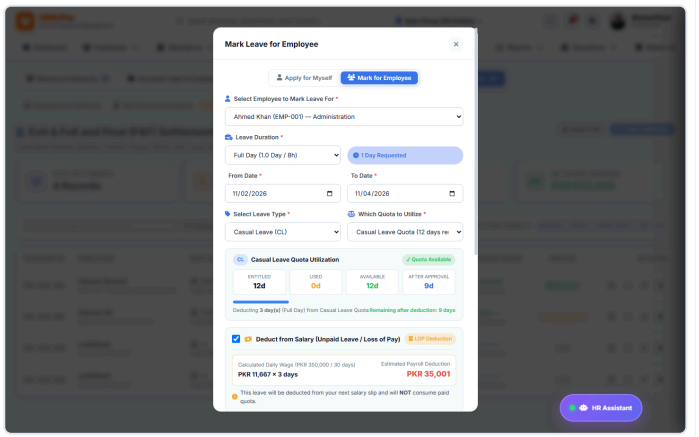


Leave Application: 3-Day Request (Scenario 5.2)



FBR Progressive Tax & Net Pay Reconciliation (Scenario 6.1)





ATS Kanban Pipeline Progression: 92% Match (Scenario 7.1)

Statutory Exit Gratuity Settlement: PKR 750k (Scenario 9.1)